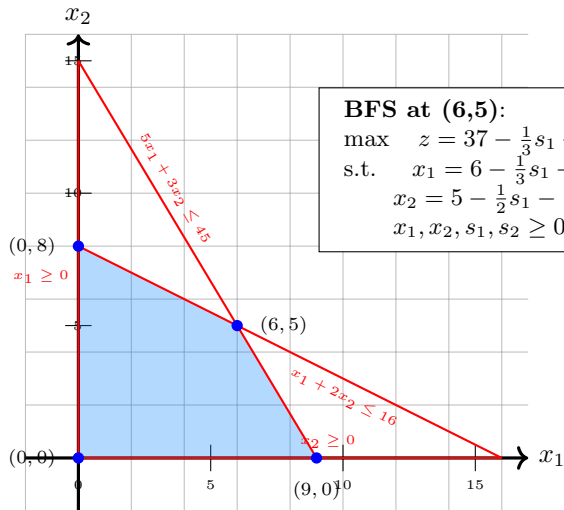


BFS at (0,8)

$$\begin{aligned} \max \quad & 2(0) + 5(8) + 2x_1 \\ \text{s.t.} \quad & x_1 + 2(8) \leq 16 \\ & 5x_1 + 3(8) \leq 45 \\ & x_1, s_2 \geq 0 \end{aligned}$$

**BFS at (6,5):**

$$\begin{aligned} \max \quad & z = 37 - \frac{1}{3}s_1 + \frac{11}{6}s_2 \\ \text{s.t.} \quad & x_1 = 6 - \frac{1}{3}s_1 + \frac{1}{3}s_2 \\ & x_2 = 5 - \frac{1}{2}s_1 - \frac{1}{2}s_2 \\ & x_1, x_2, s_1, s_2 \geq 0 \end{aligned}$$

Vertex	Objective Value
(0,0)	0
(0,8)	40
(6,5)	37
(9,0)	18

BFS at (0,0)

$$\begin{aligned} \max \quad & 2x_1 + 5x_2 \\ \text{s.t.} \quad & s_1 = 16 - x_1 - 2x_2 \\ & s_2 = 45 - 5x_1 - 3x_2 \\ & x_1, x_2 \geq 0 \end{aligned}$$

BFS at (9,0)

$$\begin{aligned} \max \quad & 2(9) + 5(0) + 5x_2 \\ \text{s.t.} \quad & 9 + 2x_2 \leq 16 \\ & 5(9) + 3x_2 \leq 45 \\ & x_2, s_1 \geq 0 \end{aligned}$$